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The Terawatt Gap: Why Compute Scarcity Is the New Macro Reality

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The conflict in the Middle East has forced most investors to factor upcoming energy-related shortages into their portfolio decisions. However, while attention has been focused on the Strait of Hormuz, a structural shortage has been building rapidly elsewhere. The early signs of a global compute shortage are no longer theoretical. They are already visible across multiple layers of the semiconductor stack and continue to spread. What began as tightness in memory, particularly high-bandwidth memory, is now extending into CPUs and broader system-level constraints. At the same time, leading AI platforms are showing signs of strain. This is increasingly evident in real-world usage. Frontier models, particularly as agentic tools are rolled out, are increasingly introducing usage limits, signaling that demand is pushing against real infrastructure ceilings. Pricing is also beginning to shift, with subtle increases and restructuring reflecting the rising cost of compute rather than simply the value of software. As a power user, I have felt this for the very first time. Meanwhile, the discussion around next-generation models such as Mythos highlights the increasing scale of what is coming next: systems that will require another step higher in compute intensity. Taken together, these developments point to a clear reality. While the world remains focused on traditional industrial inputs like oil, it is overlooking a parallel constraint forming in semiconductors. It is not preparing for a structural semiconductor shortage. It is already living through one, and the constraints are beginning to shape both user experience and the pace of AI development.

What makes these shortages even more important is that they are emerging at a very early stage of the AI buildout. The industry has only started to feel the true scale of inference demand, especially as the paradigm shifts from static cloud-based models to dynamic agent-driven systems that require continuous, real-time computation. What began as isolated

bottlenecks is broadening into a system-wide shortage across nearly every class of chip, with particularly acute pressure in advanced packaging, such as CoWoS, and in power management ICs, two critical pinch points that highlight how physical infrastructure, not just silicon design, is becoming the limiting factor. This is more than an extension of the original data center expansion cycle. It is a structural shift in how compute is consumed. AI agents, unlike traditional workloads, generate persistent, multi-step interactions that raise inference intensity and sustain pressure on CPUs, memory, networking, and specialized accelerators. Demand is no longer concentrated in a small number of hyperscale deployments. It is spreading across the full compute stack, bringing semiconductor constraints forward much earlier than most investors expected.

This backdrop is what makes Elon Musk's recent Terafab announcement so important. Forget what he is planning to do and focus on the message he is delivering to the world. His push to build a Terafab is more than an ambitious expansion plan. It is a direct response to the gap between future chip demand and the industry's current ability to supply it. Musk has openly suggested that his own needs, driven by robotics, autonomous systems, and embodied AI, will exceed what the global semiconductor ecosystem can deliver. He has also highlighted the fragility of the supply chain: one dominant advanced lithography provider and only a handful of fabrication leaders capable of operating at the frontier. In Musk's view, the compute required in the years ahead, potentially measured in terawatts, cannot be supported within this narrow framework. That is what is pushing him toward vertical integration on an unprecedented scale. The move reflects ambition, but it also reflects necessity, because the next phase of AI, centered on physical and embodied systems, demands a level of semiconductor output that the current global infrastructure was never built to provide. AI is now running into the constraints of the physical world that ultimately powers it.

The numbers attached to Terafab are what make the announcement impossible to ignore. According to Elon, current global AI chip production is framed at roughly 20 gigawatts per year, while his own future needs imply something closer to 1 terawatt of compute, suggesting his demand alone could approach 50 times today's annual industry output. The proposed Terafab itself is described at roughly 100 million square feet, a footprint compared to three Central Parks, 15 Pentagons, and more than 1,700 football fields. Just as important, this demand is not being framed around one narrow AI use case. It is tied to a future that includes humanoid robots, autonomous vehicles, and space-based compute infrastructure. Even if one discounts the most aggressive edges of the vision, the scale of the numbers tells you what matters most: Elon is signaling that the current semiconductor system is radically undersized for the world he believes is coming.

His ecosystem, spanning autonomous vehicles, humanoid robotics, and longer-term space ambitions through SpaceX, points to a level of chip consumption that far exceeds what today's fabrication base was designed to support. A future SpaceX IPO matters in this context because it will unlock enormous capital to fund a vertically integrated semiconductor buildout, effectively using the valuation of one frontier business to finance the physical infrastructure needed for another. The deeper point is strategic. Musk is trying to reduce dependence on a fragile and concentrated supply chain by bringing more of the stack under his own control, from design to manufacturing to deployment. In doing so, he is recognizing a reality that much of the market is only beginning to appreciate: the future of AI, especially embodied AI, will be constrained by the physical ability to produce enough semiconductors to power it. Regardless of one's view of Elon Musk, Terafab is a signal to the world that the mathematical, voracious appetite of AI for semiconductors cannot be met by the current infrastructure. Demand outstrips supply and according to him, the only thing that will slow semiconductor usage is the power and raw commodity needs to keep them running.

Taken together, these dynamics point to a persistent blind spot in the market's interpretation of AI. My goal is to highlight the structural realities shaping this transition, whether through specific companies or broader thematic shifts. At times, that means focusing on individual names. At other times, it means stepping back to emphasize the larger forces that are easy to miss in real time. The world has not invested over the last 17 years for the needs of the rise of AI agents and certainly not for embodied AI. This leaves investors continuing to underestimate the exponential nature of compute demand as intelligence moves through each new phase of development. Anchoring to past industrial cycles prevents investors from recognizing the nature of this shift. Many still look at parabolic charts and fail to see them for what they are: signs of a world unprepared for what Elon Musk sees coming. Whenever there is a pullback, always have levels in mind for those names you like. Commodities trade differently than software.

So far, the world has largely been in the brain-building phase, focused on raising model intelligence through pre-training. Now it is entering the digital employee phase, where AI agents can scale from thousands to potentially billions of always-

on systems performing real work. After that comes the buildout of embodied AI, where intelligence moves into robotics, autonomous systems, and machines operating continuously in the physical world. In that context, Terafab may sound extreme, but the real mistake has consistently been underestimating AI's appetite for compute. Each phase has expanded demand far beyond prior expectations, and that pattern is unlikely to change. That is why my thematic portfolio of 100 names is built around this transition. It is heavily weighted not only toward semiconductors, but also toward the infrastructure required to support today's AI buildout and the embodied systems still ahead. Notably, silver stands as the only commodity included on that list, reflecting its unique role in enabling the efficient transfer of energy required to power this next generation of compute. Elon Musk's announcement highlights that this remains an early-stage theme that should extend into edge devices and a much broader physical compute economy over the coming years. For readers interested in the specific companies exposed to this buildout, my broader research papers outline the names and infrastructure layers most leveraged to this transition. Again, long scarcity, short abundance.

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